

SQL Data Analysis Internship – Task 2

✚ Task Description

In Task-1, you created a Student Management Database and wrote basic queries. Now, let's **extend the database** by adding new tables for **Courses and Enrollments**. You'll then write queries to analyze performance by course and find insights like top students. This step will introduce you to **joins, group by, and aggregations**.

What You'll Build (Requirements)

1. Extend DB:
 - **courses** (id, name)
 - **enrollments** (student_id, course_id, grade)
2. Write SQL Queries:
 - List all students enrolled in each course.
 - Find average grade per course.
 - Find top 3 students overall.
 - Count students who failed (grade < 40).

Why This Task Is Important

- You'll learn **table relationships** (one-to-many).
- Practice **joins and aggregations**, essential for analytics.
- Start thinking like a **data analyst**, not just writing queries.

✓ Deliverable

A SQL script file with:

- Schema for new tables.
- Queries written.
- Screenshots of query results.

Sample Starter Code

Schema

sql

```
CREATE TABLE courses (  
  id INT PRIMARY KEY,  
  name VARCHAR(50)  
);  
  
CREATE TABLE enrollments (  
  student_id INT,  
  course_id INT,  
  grade INT,  
  FOREIGN KEY (student_id) REFERENCES students(id),  
  FOREIGN KEY (course_id) REFERENCES courses(id)  
);
```

Queries

sql

```
-- Students in each course  
SELECT s.name, c.name  
FROM enrollments e  
JOIN students s ON e.student_id = s.id  
JOIN courses c ON e.course_id = c.id;  
  
-- Average grade per course  
SELECT c.name, AVG(e.grade) as avg_grade  
FROM enrollments e  
JOIN courses c ON e.course_id = c.id  
GROUP BY c.name;  
  
-- Top 3 students  
SELECT s.name, AVG(e.grade) as avg_grade  
FROM enrollments e  
JOIN students s ON e.student_id = s.id  
GROUP BY s.name  
ORDER BY avg_grade DESC  
LIMIT 3;
```

Helpful Resources

- [SQL Joins Tutorial – W3Schools](#)
- [SQLBolt Lessons](#)
- [Mode Analytics SQL Tutorial](#)

✂ Tools

- MySQL / PostgreSQL
- DB Fiddle (online SQL runner)
- VS Code (for .sql files)